

Wk1: Distribution Theory and Properties of Stock Price Returns I

Nicky Grant (Semester 1, 2021/2022)

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Course Details

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Office Hours: Online Monday 15:10-17:00 (or by appointment)

Lecture Topics

- 1 Distribution Theory and Properties of Stock Price Returns I
- 2 Distribution Theory and Properties of Stock Price Returns II

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- 6 Maximum Likelihood & GARCH Models
- 8 Advanced Volatility Modelling
- 9 Unit Roots & Co-integration

30% Empirical Project & 20% Class Test

Assessment and Reading

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50% Final Examination

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Main Textbooks:

Hurn and Martin: Financial Econometric Modeling (**HM**)

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Greene: Econometric Analysis (**G**)

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Supplementary Textbooks:

Linton: Financial Econometrics: models and methods (**L**)

Financial Econometric Data Examples

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See **HM** Chapter 1 for more general definitions taking into account **dividends** and also **continuously compounded** and **k-period ahead returns**

Example Financial Return Data

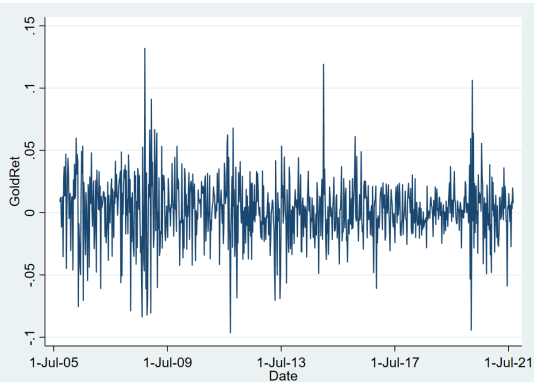


Figure 1: Weekly Gold Price Return
2005-2021

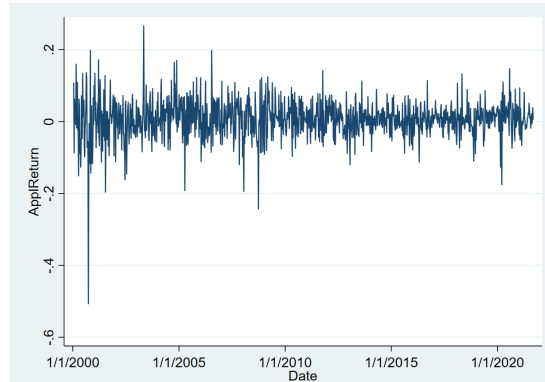


Figure 2: Weekly Apple Stock Return
2000-2021

Example Financial Return Empirical Densities

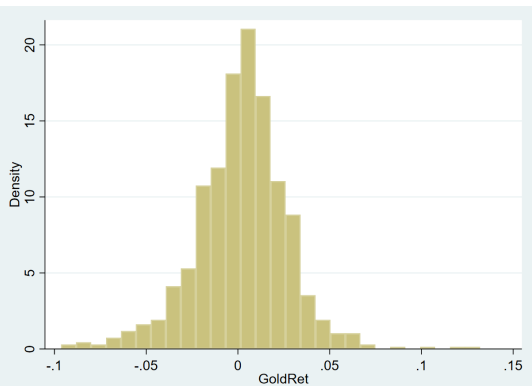


Figure 3: Empirical Density Weekly Gold Return 2005-2021

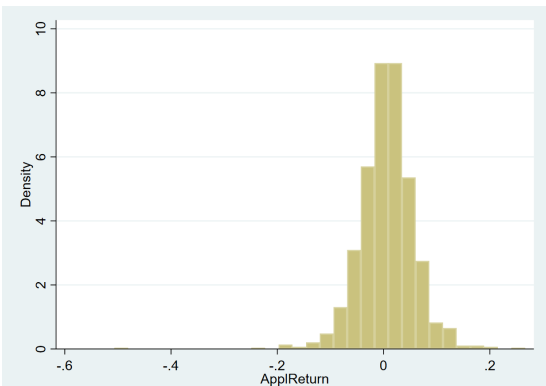


Figure 4: Empirical Density Weekly Apple Stock Return 2000-2021

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Volatility Clustering

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Volatility Clustering

Slow decay in correlation is absolute returns

Statistical Distribution Theory and Definitions

Motivation

Often interested in modelling the distribution of financial economic data

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E.g. the distribution of stock returns, or volatility, interest rates, exchange rates...

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Crucial to understand key definitions and results on statistical distribution theory and sampling

Discrete Variables: Univariate Probability Density and Distribution Functions

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Satisfies axioms of probability

Discrete Variables: Example- Bernoulli

Takes two values: $\text{Supp}(X) = \{0, 1\}$

$$f_X(x) = \begin{cases} p, & x = 1, \\ 1 - p, & x = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Continuous Variables: Univariate Probability Density and Distribution Functions

A continuous r.v. X takes values over a continuous interval, e.g. $\text{Supp}(X) = \mathbb{R}$

Probability distribution function: $F_X(x) := \int_{-\infty}^x f_X(x)dx$

$f_X(x)$ is the **probability density function** at $X = x$

Continuous Variables: Example- Standard Uniform Distribution Function

Constant density function with $\text{Supp}(X) = [0, 1]$

$$f_X(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Continuous Variables: Example- Standard Distribution Function

Bell-shaped density function with $\text{Supp}(X) = \mathbb{R}$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$$

Definition: Population Moments of X

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Var of X $\mathbb{E}[(X - \mu_X)^2]$

$$\sigma_X^2 := \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx$$

Skewness of X : $\mathbb{E} \left[\left(\frac{X - \mu_x}{\sigma_X} \right)^3 \right]$

Kurtosis of X : $\mathbb{E} \left[\left(\frac{X - \mu_x}{\sigma_X} \right)^4 \right]$

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We give definition for bivariate r.v.s [$k = 2$] (but definition generalises- see)

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Marginal density $f_{X_1}(x_1) = \sum_{x_2} f_X(x_1, x_2)$

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$$\text{Cov}(X_1, X_2) := \mathbb{E}[(X_1 - \mu_{X_1})(X_2 - \mu_{X_2})] \quad \textbf{Covariance } X_1, X_2$$

$$\text{Corr}(X_1, X_2) := \text{Cov}(X_1, X_2) / \sigma_{X_1} \sigma_{X_2} \quad \textbf{Correlation } X_1, X_2$$

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$f_{X_1|X_2=x_2}(x_1)$ is a probability function- can use to work out probability statements on X_1 , conditioning on the event $X_2 = x_2$

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$(x_1, \dots, x_T)'$ is observed sample data from $\{X_1, \dots, X_T\}$

Sampling and Moment Estimation

Sampling: i.i.d assumption

Would like to use observed data to estimate the distribution(or moments) generating the data

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Generally does not hold with time series data, where modelling the dependence is the main interest

Definition: Weakly Stationary Process

A process $\{Z_t : t \in \mathcal{T}\}$ is **weakly stationary** if

$$\mathbb{E}[Z_t] = \mu \quad \text{for all } t \in \mathcal{T},$$

$$\text{Var}(Z_t) = \sigma^2 < \infty \quad \text{for all } t \in \mathcal{T},$$

$$\text{Cov}(Z_{t_1}, Z_{t_2}) = \gamma(|t_1 - t_2|) \quad \text{for all } t_1, t_2 \in \mathcal{T}.$$

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Definition: Strictly Stationary Process

A process $\{Z_t\}$ is **strictly stationary** if for all $t_1, \dots, t_k$ in $\mathcal{T}$ the joint density of $(Z_{t_1+\tau}, \dots, Z_{t_k+\tau})'$ does not depend on τ .

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Z_t has a unit root (e.g a random walk)

Definition: White Noise

A process $\{Z_t\}$ is **white noise** if

$$\mathbb{E}[Z_t] = 0 \quad \text{for all } t \in \mathcal{T},$$

$$\mathbb{E}[Z_t^2] = \sigma^2 < \infty \quad \text{for all } t \in \mathcal{T},$$

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Note: white noise $\Rightarrow$ weak stationarity (not the reverse)

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Correlations in sample data should reflect the correlation in X_t (for large T)

Assuming DGP is (weakly) stationary, under some mild conditions, sample averages of moments consistently estimate the population moment.

Sample Mean: $\bar{\mathbf{X}}_T = \frac{1}{T} \sum_{t=1}^T X_t$ consistently estimates μ_X

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Likewise can construct time averages of variance, skewness and kurtosis

Empirical Example Cont...

GoldRet				
Percentiles		Smallest		
1%	<u>-.0705071</u>	<u>-.0963523</u>		
5%	<u>-.0411961</u>	<u>-.0942303</u>		
10%	<u>-.0283615</u>	<u>-.0837025</u>	Obs	833
25%	<u>-.0115863</u>	<u>-.0821838</u>	Sum of wgt.	833
50%	<u>.0032998</u>		Mean	<u>.0019841</u>
		Largest	Std. dev.	<u>.0253551</u>
75%	<u>.0160616</u>	<u>.0911392</u>		
90%	<u>.0303873</u>	<u>.1063008</u>	Variance	<u>.0006429</u>
95%	<u>.0389286</u>	<u>.1191333</u>	Skewness	<u>-.0784091</u>
99%	<u>.0638143</u>	<u>.1319216</u>	Kurtosis	<u>5.432574</u>

Figure 5: Sample Moment Summary of Weekly Gold Price Return 2005-2021

ApplReturn				
Percentiles		Smallest		
1%	<u>-.1267848</u>	<u>-.5065874</u>		
5%	<u>-.0734749</u>	<u>-.2430599</u>		
10%	<u>-.0521449</u>	<u>-.195976</u>	Obs	1,130
25%	<u>-.0217912</u>	<u>-.1942862</u>	Sum of wgt.	1,130
50%	<u>.0076731</u>		Mean	<u>.0061534</u>
		Largest	Std. dev.	<u>.052576</u>
75%	<u>.0355935</u>	<u>.1719746</u>		
90%	<u>.0661697</u>	<u>.1983422</u>	Variance	<u>.0027642</u>
95%	<u>.0830427</u>	<u>.1986522</u>	Skewness	<u>-.8683249</u>
99%	<u>.1336532</u>	<u>.2664362</u>	Kurtosis	<u>12.41757</u>

Figure 6: Sample Moment Summary of Weekly Apple Stock Return 2000-2021

Definition: Sample Auto-Covariance Function

$$\hat{\gamma}_T(k) = \frac{1}{T-k} \sum_{t=k+1}^T (x_t - \bar{x}_T)(x_{t-k} - \bar{x}_T) \quad \forall |k| < T$$

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Definition: Sample Auto-Correlation Function

$$\hat{\rho}_T(k) = \frac{\hat{\gamma}_T(k)}{\hat{\gamma}_T(0)} \quad \forall |k| < T$$

Definition: Sample Auto-Covariance Function

$$\hat{\gamma}_T(k) = \frac{1}{T-k} \sum_{t=k+1}^T (x_t - \bar{x}_T)(x_{t-k} - \bar{x}_T) \quad \forall |k| < T$$

Definition: Sample Auto-Correlation Function

$$\hat{\rho}_T(k) = \frac{\hat{\gamma}_T(k)}{\hat{\gamma}_T(0)} \quad \forall |k| < T$$

Under stationarity $\hat{\gamma}_T(k)$ consistently estimates $\gamma(k)$ as $T \rightarrow \infty$

Empirical Example Cont...

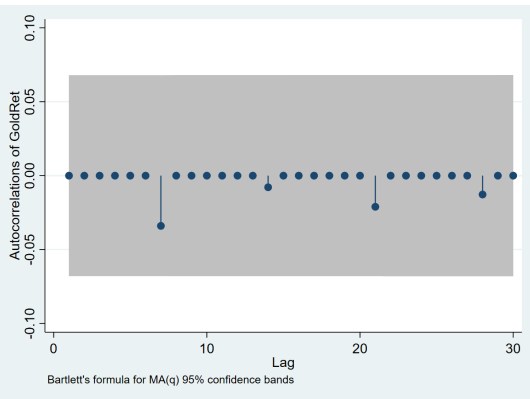


Figure 7: Sample Correlogram of Weekly Gold Price Return 2005-2021

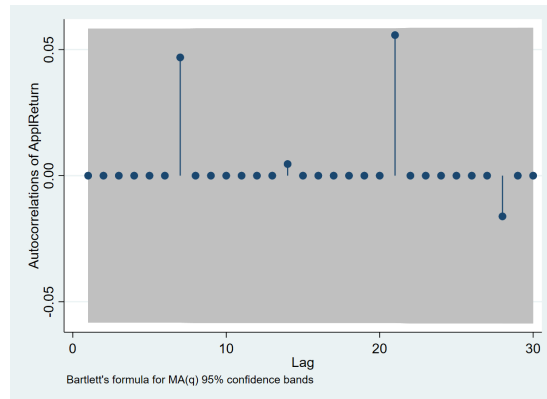


Figure 8: Sample Correlogram of Weekly Apple Stock Return 2000-2021

B Chapter 1.1-1.5 [*A very gentle introduction to financial econometrics*]

B Appendix A1-A5 and **G** Appendix A1 and A2 and B1-B5 [Recap of some basic maths and stats results- **B** is a more basic introduction, those more advanced may skip to **G**]

H Chapters 1 and 2 [A more thorough intro to financial econometrics and properties of financial returns, leading onto next Week's material]

Some further discussion on the first four moments- mean, variance/standard deviation, skewness and kurtosis: [http://rovdnloads.com/attachments/newsletters/Newsletter%2003%20-%20Understanding%20the%20Forecast%20Statistics%20and%20Four%20Moments%20\(4P\).pdf](http://rovdnloads.com/attachments/newsletters/Newsletter%2003%20-%20Understanding%20the%20Forecast%20Statistics%20and%20Four%20Moments%20(4P).pdf)